[ANSWER KEY] Total Points: [10] [U1HW8 - W11 - Vector Kinematics and Dynamics 1]

1. The study of why objects move, examining how forces cause objects to accelerate, slow down, or turn.

2. Vector kinematics describes how objects move using directional measurements like displacement and velocity without focusing on the forces causing motion.

3. Newton (N)

4. The rotating paddle blades push backward against the water, and the water exerts an equal forward reaction force on the boat.

5. Directly opposite to the direction of the hull's forward motion.

6. The numerical size, amount, or physical quantity expressed by a number and unit, independent of direction.

7. 18 Newtons (or 18 N)

8. By the physical length of the arrow.

9. 5 times greater

10. It has magnitude only and is completely described by a number and a unit without direction.